

Course outline Winter 2025

Course Name: Android Development

Course Code: CS-3130

INSTRUCTOR

Instructor(s)	Dr. Hossam Mekawy
E-mail	Hossam.mekawy@uofcanada.edu.eg
Office Location	B211
Office Hours	(by appointment – please send an email a day before to check if the time slot is not already reserved) Sunday 8:30 AM-9:30 AM Thursday: 2:55 PM – 3:55 PM

TEACHING ASSISTANT

Instructor(s)	Eng. Omar Eltouny
E-mail	Omar.Aymen@uofcanada.edu.eg
Office Location	B318
Office Hours	Wednesday: 2-3 PM Thursday: 11 AM-12 PM

Class Time

Lecture	Lab
Monday 8:30 am - 09:45 pm	Tuesday 8:30 – 9:45 am
Monday 3:00 pm - 4:15 pm	

Class LOCATION

Lecture	Lab
B102, A513	A513

COURSE DESCRIPTION

This course introduces the student to programming for mobile devices that use the Android platform. The course will present a study of the architecture, operating system, and programming language of these devices. Three lecture hours per week

Requisites:

Students should have successfully completed Computer Science III and Data Structures.

RECOMMENDED TEXTBOOKS

Griffiths, D., 2019, "Head First Kotlin: A Brain-Friendly Guide", O'Reilly Media.

Griffiths, D., & Griffiths, D., 2021, "Head First Android Development: A Learner's Guide to Building Android Apps with Kotlin", O'Reilly Media.

Software

Java JDK 19

Kotlin

Android Studio

Git, GitHub

LEARNING OBJECTIVES

On completion of this course, students should be able to:

LO1. Demonstrate a strong grasp of the Kotlin programming language.

LO2. Structure an Android project in accordance with best practices.

LO3. Design dynamic, interactive Android user interfaces.

LO4: Independently seek knowledge to implement Android views and use APIs.

LO5: Manage and oversee an Android project from ideation to release.

CLASS SCHEDULE AND GRADING SCHEME

Week	Lecture	Lab
1 <i>Feb 3rd</i>	Introduction - Course introduction, - Android studio setup and IDE Participation - Tools installation - 1st project	- Tools installation - 1st project.

2 Feb 10th	Introduction - introduction to Kotlin, UI Components: - Layouts - o Linear layout - o Relative layout - o Table layout - o Constraint layout Participation - Create calculator class using Kotlin - Utilize github	- Create calculator class using Kotlin - Utilize github
3 Feb 17th	Views - Views - o Text view - o Button - o Highlights of edit text, image view, check box, radio buttons, and spinners User interaction - Toast message - Snackbar message - Dialog Participation Create a registration screen	Build a registration screen
4 Feb 24th	Lists and Views - List view - Recycler View Participation: A Task involves:	A Task involves: - Alert dialog

	- Alert dialog	
5 <i>March 3rd</i>	Intents, lifecycles, and services - Intent - Activity lifecycle - Services Participation - Multilingual support - DPI support - Different screens support	- Multilingual support - DPI support - Different screens support
6 <i>March 10th</i>	Midterm	
7 <i>March 17th</i>	Shared preferences, view binding - Shared Preferences - View binding - Broadcast receiver Participation - Persistence, versioning, & publishing	Persistence, versioning, & publishing
8 <i>March 24th</i> <i>online</i>	Broadcast receivers - Broadcast receiver: the manifest way - Broadcast receiver: register receiver Participation - Navigation	Navigation
10 <i>April 7th</i>	Fragments - Creating fragments - Replacing fragments - Fragments backstack - List fragment Participation - DialogFragment & Common Intent	DialogFragment & Common Intent
11	local notifications, REST API	REST API exercise

	- Local notification - Retrofit library - Data retrieval Participation REST API exercise	
11 April 14th	Firestore - Adv./Disadv. Firestore - Products of interest - Real time Database - Push notification Participation - Firestore authentication	Firestore authentication
12 April 21st	Project	Project milestone1 support
13 April 28th	Project	Project milestone 2 support

GRADING SCHEME

Assessments	Grade	Date
<i>Lecture participation</i>	10%	All lectures
<i>Quiz1</i>	2.5%	Week 3: Feb 17 th
<i>Quiz 2</i>	2.5%	Week 5: March 3 rd
<i>Quiz 3</i>	5%	Week 10: April 7 th
<i>Assignment 1</i>	7%	Week 3: Feb 17 th
<i>Midterm</i>	15%	Week 6: March 10 th
<i>Assignment 2</i>	7%	Week 6: March 10 th
<i>Assignment 3</i>	7%	Week 10: April 7 th

Project	44%	Week 13: April 28 th
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QUIZZES

Quiz	topics	Sub-weighting
1	Kotlin, Android Basics, Android components	2.5%
2	User interactions, List and views, intents and life cycles	2.5%
3	Save/Load data, shared preferences, device compatibility, fragments, sending data between screens. notifications, Firebase. REST API	5%

HOMEWORK/ASSIGNMENTS

- You are expected to complete all weekly reading in preparation of class. Occasionally I may also hand out articles or cases not in the text that are required for you to read and prepare for class discussion. In addition, you will have individual/group assignments and term-based reports and assignments throughout the term. **ALL REQUIRED COURSE WORK MUST BE SUBMITTED TO PASS THE COURSE.**
- All reports required during this course shall be original and follow academic integrity policy. The reports will be submitted on Turnitin tool, Turnitin check for similarities among other reports and the internet. If similarity beyond reasonable amount is detected penalties extending to filing an academic dishonesty case may be applied
- All students are expected to submit their work on time, late submissions without prior approved excuse will not be accepted.
- Students are required to use report templates if posted by the instructor.
- This course has lectures, assignments, a project and exams.
- Each assignment is a small programming project that will cover topics presented in lectures before the due date, as well as extra research tasks that the student is expected to autonomously conduct. It is expected that each assignment requires approximately 10 hours of effort, so time management is paramount to ensure they are completed before the due date.
- Students will be provided with a specification that gives information about what is expected from each assignment and how marks are allocated.
- Failure to adhere to the academic misconduct expectations outlined below will result in a zero mark.
- The student is expected to finish and deliver an answer to the hands-on exercises initiated in each lecture by the next lecture. Participation grades are dependent on such a delivery. Office hours and optional labs are available to provide support as needed.
- Bonus grades are typically attached to some activities and cannot exceed a total of 5%.

EXAMS

As per the assessment dates table, generally, unless otherwise stated by the instructor, midterm exam/s (if applicable) will be held during the regularly scheduled lecture time.

Starting from week 3, tests will occur every fortnight and will cover topics presented in the previous weeks. These will be completed during the second lecture of the week. Each quiz will take approximately 15 minutes and will feature multiple choice questions.

ATTENDANCE AND PUNCTUALITY

Students in this course are required to attend all classes and be on time for all classes. There are no excuses for missing class. All absences are counted as time missed. This applies to all lectures, practical tutorials and/or labs. This also applies to mandatory office hours for probation & Gateway students. Students miss out on opportunities to acquire academic skills such as critical thinking, problem-solving, oral communication, teamwork, and decision-making when they fail to attend class. Being late for class disrupts the flow of the lecture and is disrespectful to the instructors and fellow students. Accordingly, students should note the following:

1. The instructors are not required to give substitute assignments or make up examinations for students who are absent.
2. It is the student's responsibility to make up the class content missed in consultation with his/her instructor(s). This does not mean that the instructor will provide a substitute class for the student.
3. Grades can be affected by missing a class as the instructor may include class participation marks and/or in-class assignments, or pop-up quizzes as part of the assessment plan for the course.
4. If a student is absent for 25% or more of the class time, tutorials, labs, and/or mandatory office hours, the instructor may:
 - a. Advise the student to drop the course if the absences occur before the drop date for course.
 - b. Give the student a grade of F for the course
 - c. Give the student an Incomplete grade for the course if the absences are caused by an approved documented illness or emergency. Students would then be given the opportunity to complete the course work as per the academic regulation.

ACADEMIC INTEGRITY

All students are expected to conduct themselves in all aspects of this course with the highest level of academic integrity. Accordingly, academic dishonesty will not be tolerated. Regulation 20 of the University Calendar defines academic dishonesty. Where there is reasonable evidence to support an allegation of academic dishonesty, Regulation 20 will be strictly enforced. All students are strongly advised to become familiar with this Regulation and UPEI Cairo Academic Integrity policy which can be found at:  [UPEI Cairo academic integrity policy.pdf](#)

USE OF AI

The use of generative artificial intelligence (AI) and AI-assisted technologies is permitted only **upon receiving a written approval form the instructor**. Students should disclose in their **assignments** the use of AI and AI-assisted technologies in the writing process. Failure to do so will be considered as violation of the academic integrity policy.